DOCUMENT 10

NGW Lab Guide

Developer tools — API reference, CLI, gold sets, candidates, and workbench

NO GUESSWORK LIGHTING
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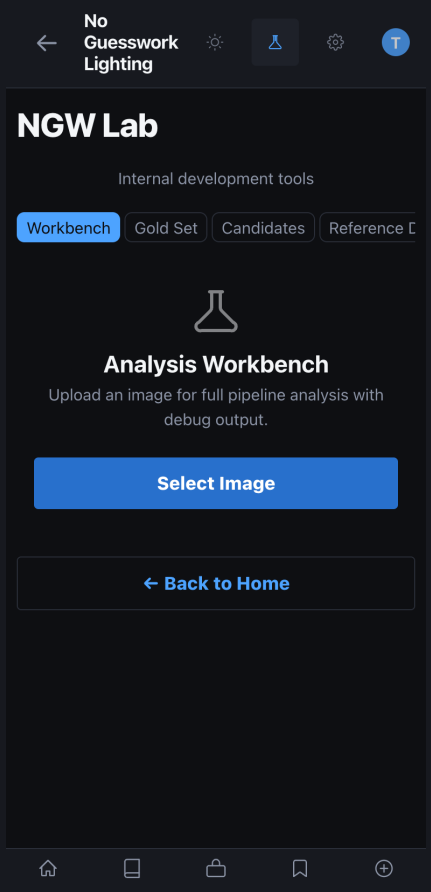
NGW

What is the NGW Lab?

The Lab is a restricted developer environment for curating ground-truth test images, reviewing engine improvement candidates, running full-fidelity analyses with debug overlays, and managing the learning signal pipeline. Access is controlled by an email allowlist (LAB_ALLOWED_EMAILS env var).

ACCESS CONTROL

• All Lab endpoints check the x-lab-email request header against LAB_ALLOWED_EMAILS. Unauthorized access returns 403 and is logged. Enable Lab UI in Settings Developer Tools.



NGW Lab Workbench — empty state showing tabs: Workbench, Gold Sets, Candidates, Signals

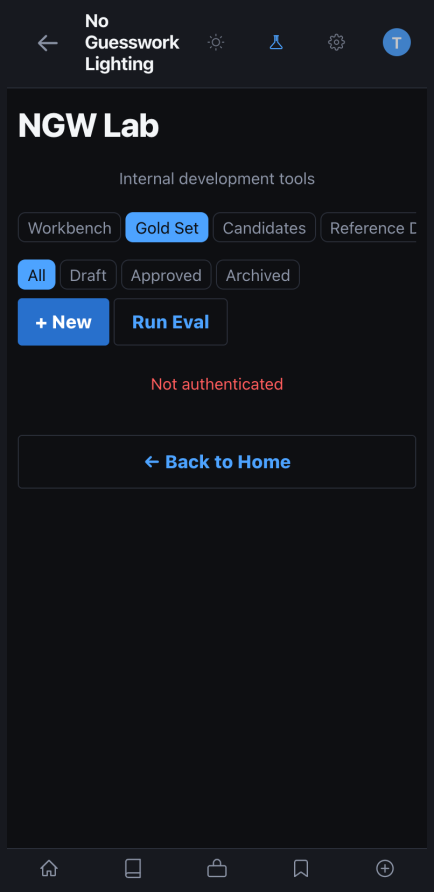
Lab API Reference — Gold Sets

POST /api/lab/gold-set — create a new gold set

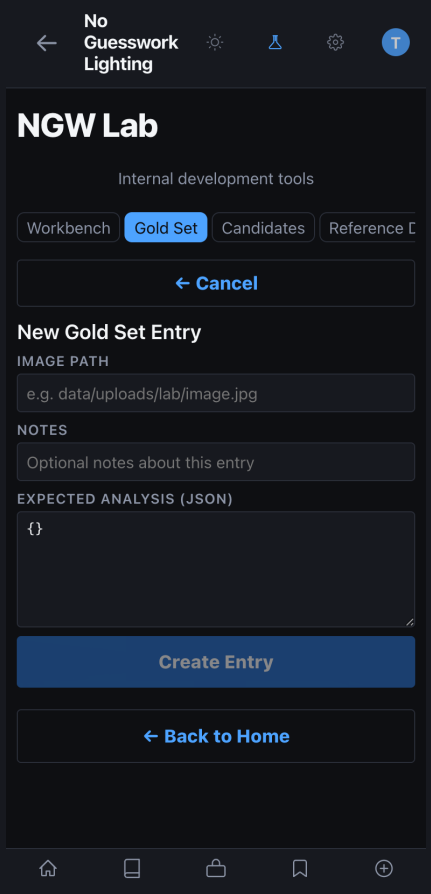
```
# Request body
{
  "image_path":    "static/uploads/portrait_001.jpg",
  "pattern":       "loop",      # curator-verified ground truth
  "subject_type":  "portrait",
  "environment":   "studio",
  "setup_details": {
    "key_light":  "Godox AD600 + octa 90cm",
    "key_angle":  45,
    "key_height": 7.5,
    "key_dist_ft": 6,
    "fill":       "V-flat reflector camera-left"
  },
  "outcome_labels": {
    "expected_pattern":    "loop",
    "expected_confidence": 0.85
  },
  "curator_email": "you@org.com"
}
# Response: {"id":"gs_uuid","status":"draft"}
```

POST /api/lab/gold-set/evaluate — run engine on approved sets

```
# Request body (optional: limit to specific IDs)
{
  "gold_set_ids": ["gs_001","gs_002"],  # omit for all approved
  "include_debug_overlay": true
}

# Response
{
  "evaluated": 12,
  "passed": 10,
  "failed": 2,
  "scores": {
    "gs_001": {"detected":"loop","expected":"loop","match":true,"conf":0.88},
    "gs_002": {"detected":"broad","expected":"loop","match":false,"conf":0.72}
  }
}
```



Gold Set listing — all test images with pattern labels, status, and evaluation scores



New Gold Set form — image upload, pattern selection, and setup metadata entry

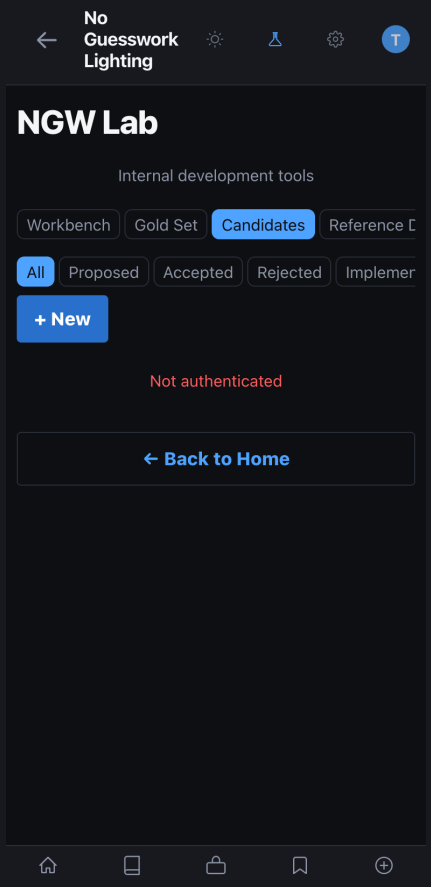
Lab API Reference — Candidates

POST /api/lab/candidates — create a candidate manually

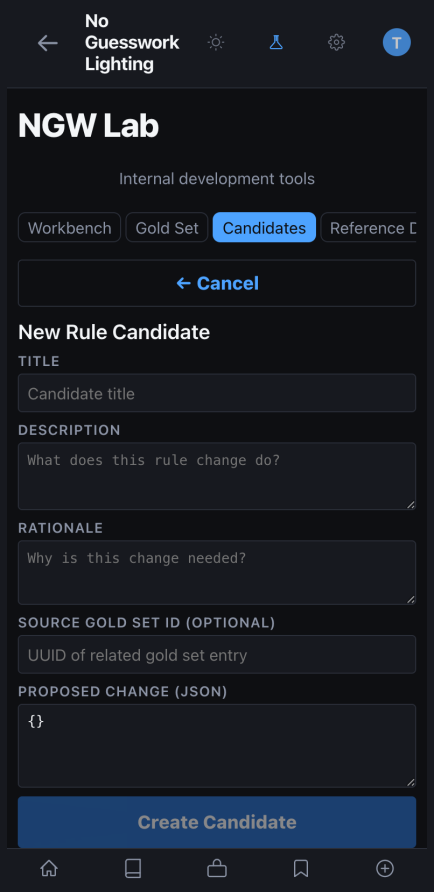
```
{
  "title":       "Recalibrate loop confidence threshold",
  "description": "Loop pattern confidence is 20% above measured CVR.
    Threshold should be lowered to reduce over-detection.",
  "candidate_type": "confidence_recalibration",
  "proposed_change": {
    "type":    "confidence_recalibration",
    "pattern": "loop",
    "action":  "lower_threshold",
    "delta":   -0.08
  },
  "estimated_success_lift":    0.12,
  "estimated_regression_risk": 0.05,
  "source": "manual"
}
# Response: {"id":"cand_uuid","status":"proposed"}
```

PUT /api/lab/candidates/{id} — update status (approve/reject)

```
{
  "status": "approved",
  "reviewer_email": "curator@org.com",
  "review_notes": "Risk acceptable. CVR data supports the change.",
  "second_reviewer_email": null
  # Required if estimated_regression_risk > 0.20
}
```



Candidates listing — proposed and reviewed engine changes with risk and lift estimates



New Candidate form — type, description, proposed change JSON, and risk estimation

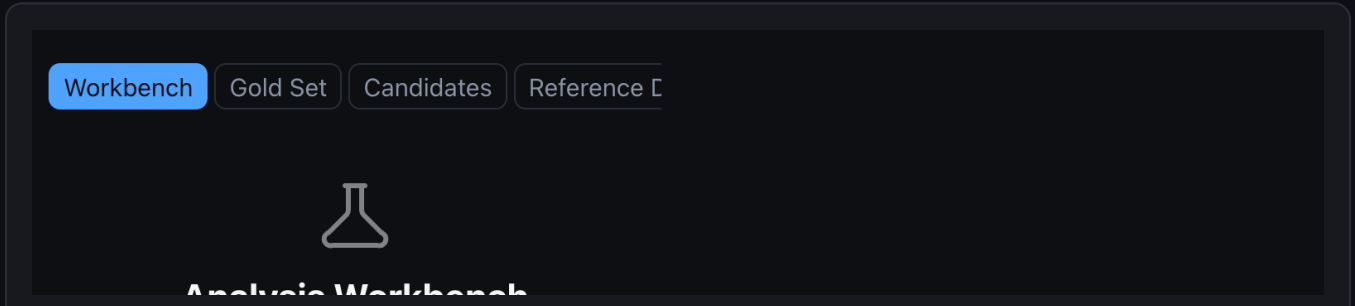
Lab API Reference — Analysis & Signals

POST /api/lab/analyze — full-fidelity debug analysis

```
# Request body
{
  "image_path":          "static/uploads/ref_001.jpg",
  "include_debug_overlay": true,
  "return_pass_details":   true,
  "kit":                   null  # null = use default test kit
}

# Response includes:
# - all three VLM pass results (raw)
# - consensus resolver scores per attribute
# - region extraction bounding boxes
# - blueprint output + gear tier
# - debug_overlay_url (annotated image)
```

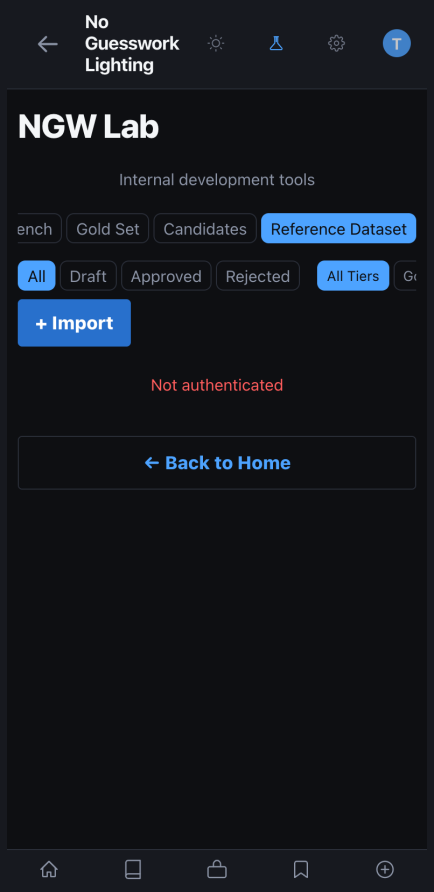
GET /api/lab/signals/summary — hygiene counts

```
# Response
{
  "live":              1482,
  "seeded":             500,
  "internal":            23,
  "expert_review":        8,
  "total":             2013,
  "learning_eligible": 1482,
  "metrics_eligible":  1482,
  "flag_mismatches":      0
}
```

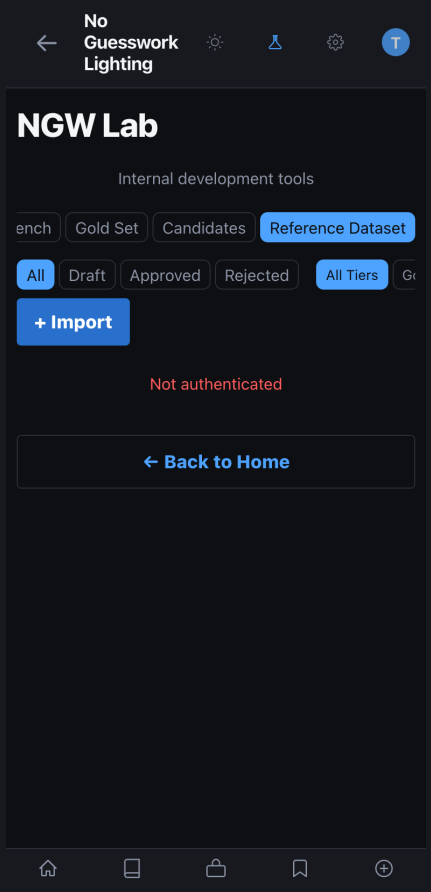


Lab — Signal Hygiene card showing live / seeded / internal / learning-eligible counts

Reference Dataset



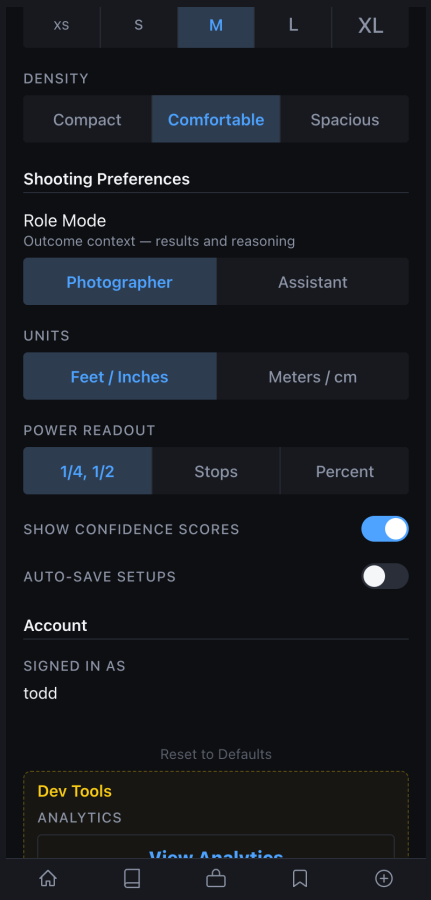
Reference Dataset — full grid view with filter bar



Reference Dataset detail — image with pattern label and approval status

POST /api/reference-library/ingest — ingest with metadata

```
{
  "image_path":   "static/uploads/ref_new.jpg",
  "pattern":      "rembrandt",
  "modifiers":    ["beauty dish","grid"],
  "subject_type": "portrait",
  "environment":  "studio",
  "quality_score": 0.92,
  "submitted_by": "curator@org.com"
}
# Starts as status="pending_review".
# Second curator approves → status="approved" → enters training set.
```



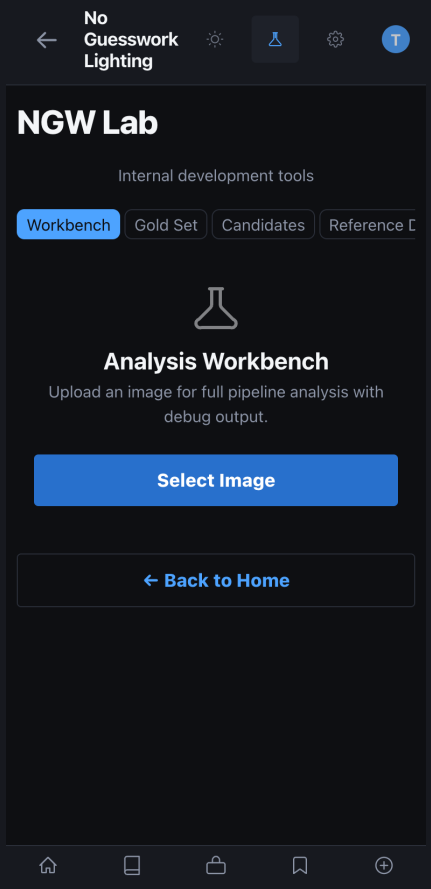
Lab Settings — evaluation thresholds, access list, and confidence gates

CLI Tools

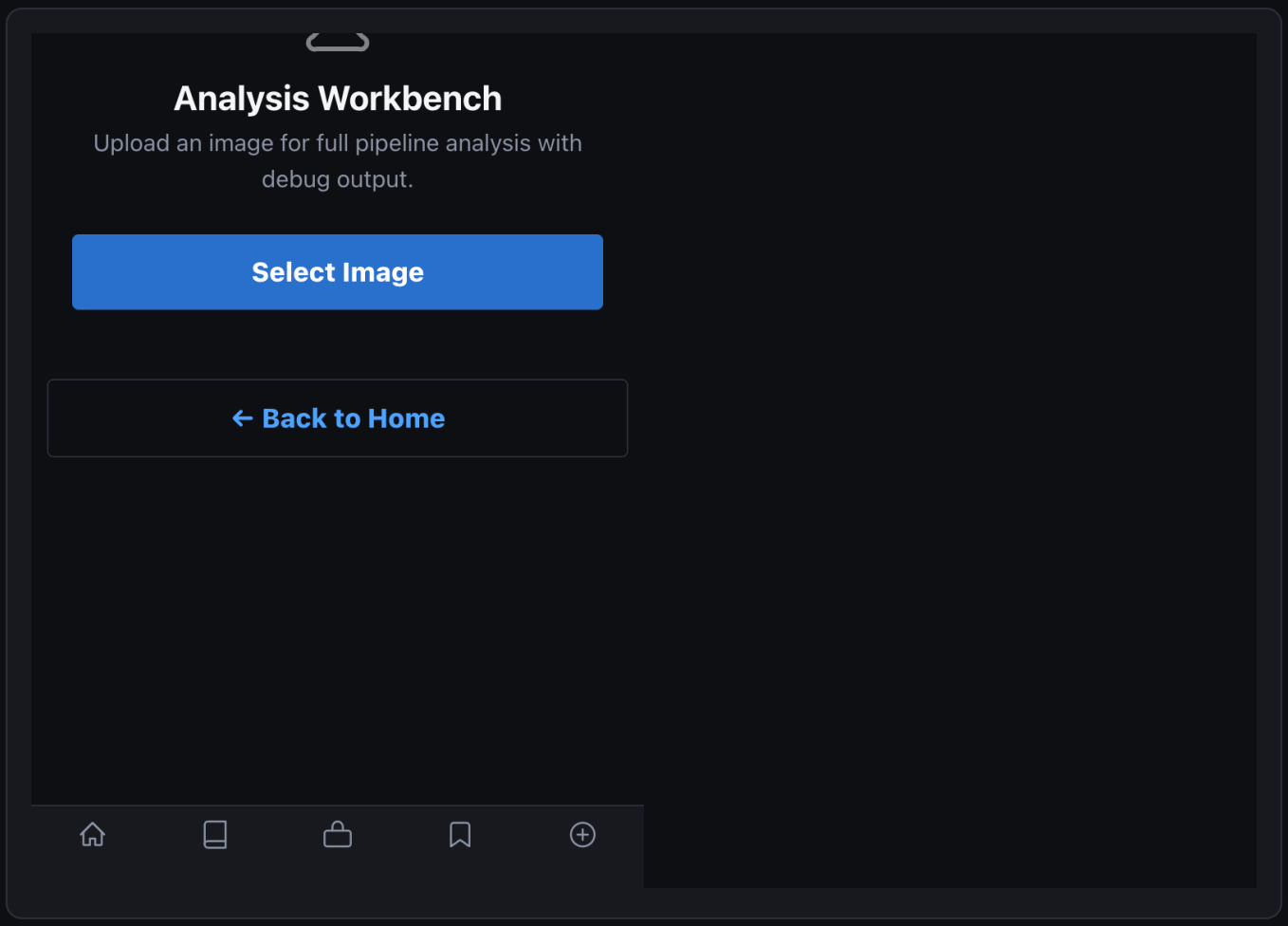
Script	Purpose	Key Flags
seed_starter_dataset.py	Load seeded signals + gold sets	--reset to clear first
verify_dataset.py	Check dataset integrity and counts	--verbose for row details
run_benchmarks.py	Pattern matching accuracy benchmarks	--gold-sets-only, --pattern=X
nightly_benchmark.py	Full benchmark suite (CI use)	--output=json
validate_system_yamls.py	Lint pattern YAML configurations	--strict
validate_catalog_and_packs.py	Validate gear catalog completeness	—
ci_benchmark.sh	CI/CD benchmark wrapper with exit codes	EXIT 1 on regression
capture_screenshots.py	Capture app screenshots for docs	--output-dir=docs/screenshots
generate_ngw_docs.py	Build this PDF documentation suite	—

bash — common Lab CLI operations

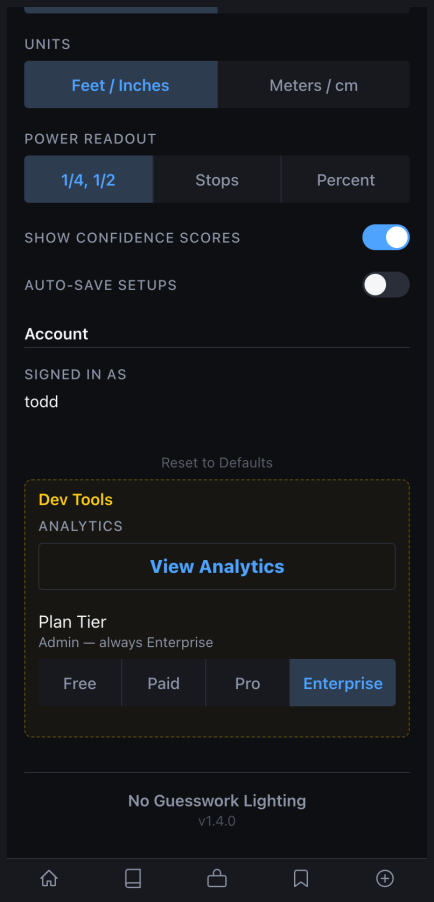
```bash
# Seed starter data (run once after init_db)
$ python3 scripts/seed_starter_dataset.py

# Verify dataset integrity
$ python3 scripts/verify_dataset.py --verbose

# Run all benchmarks
$ python3 scripts/run_benchmarks.py

# Run gold-set-only evaluation
$ python3 scripts/run_benchmarks.py --gold-sets-only

# Validate YAML configs (run before any blueprint changes)
$ python3 scripts/validate_system_yamls.py --strict
```



Workbench ready state — analysis loaded with debug overlay and region annotations



Workbench detail — per-region confidence scores and VLM pass breakdown



Dev Tools panel — database status, migration runner, and signal count monitor

How To: Add a Gold Set Image

Gold Set images are the ground-truth test set that benchmarks run against. Adding well-labeled images directly improves evaluation coverage.

1. **Open NGW Lab**
   Tap the Lab tab in the navigation bar (requires Lab access enabled).

2. **Go to Gold Sets**
   Tap "Gold Sets" in the Lab navigation tabs.

3. **Tap + Add New**
   Opens the gold set creation form.

4. **Upload the image**
   Choose a clean, unedited reference photo with clear lighting.

5. **Select pattern**
   Choose the ground-truth pattern from the dropdown (e.g. "loop").

6. **Set metadata**
   Fill in subject type, environment, modifier list, and quality notes.

7 Submit for review

Status starts as "pending_review". A second curator approves it.

8 After approval

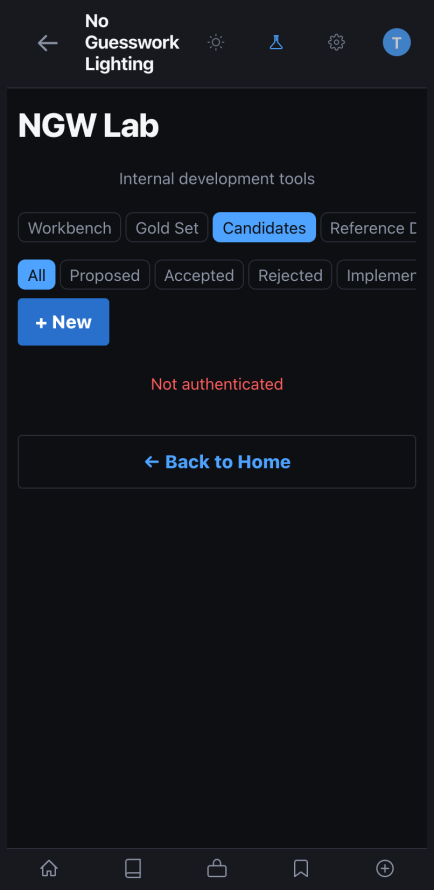
Status changes to "approved" — the image enters the benchmark set.

QUALITY STANDARDS

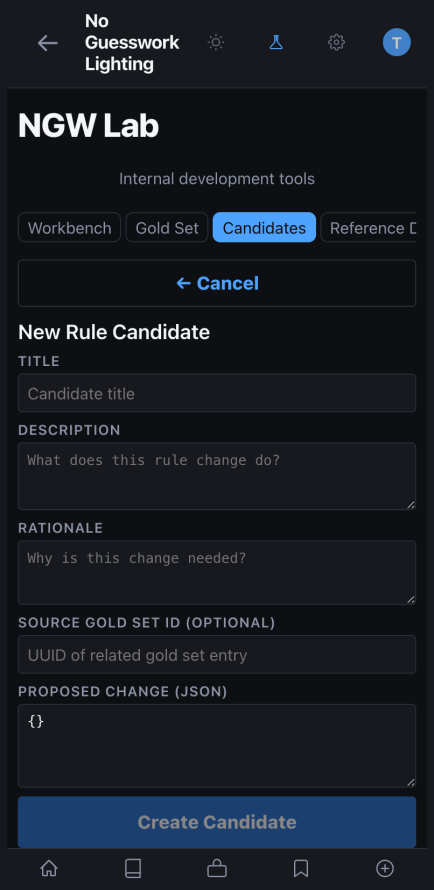
- Use only clean originals — no heavy color grading, compositing, or retouching.
- Minimum 1024 px wide. Portrait orientation preferred for face-priority patterns.
- Label with the ground-truth pattern, not what you think NGW will detect.
- Add notes for any unusual conditions (high ceiling, very large modifier, etc.).

How To: Review and Approve a Candidate

Candidates are improvement proposals — manually created or auto-generated. Review each one carefully before approving. High-regression-risk candidates require a second curator.



Candidate queue — proposed, approved, and rejected status columns



New candidate form — type selection, description, and risk assessment fields

1 Open Candidates tab

Tap "Candidates" in the Lab navigation. Filter by status="proposed".

2 Read the description

Understand what change is proposed and why the failure cluster triggered it.

3 Check regression risk

If regression_risk > 0.20, a second curator approval is required.

4 Review proposed_change

Read the full JSON. Understand which blueprint or config key changes.

5 Approve or Reject

Tap Approve to advance to "approved", or Reject to archive with a note.

6 Validate after apply

Gold set benchmarks re-run automatically. Check scores in CLI output.

NEVER SKIP VALIDATION

• Every approved candidate triggers an automatic gold set re-evaluation. If any gold set drops below 75%, the deployment pipeline fails and the candidate reverts to "proposed". This gate cannot be bypassed in production.